$$x_{CM} = \frac{(3\times 0)+(5\times 4)}{3+5}$$

$$x_{CM} = \frac{0+20}{8} = \frac{20}{8} = 2.5\,m$$

The centre of mass is located 2.5 m from 3 kg point mass, (and 1.5 m from the 5 kg point mass) on X-axis.

- This result shows that the centre of mass is located closer to larger mass.
- If the origin is shifted to the centre of mass, then the principle of moments holds good. $m_1x_1 = m_2x_2$; $3\times 2.5 = 5\times 1.5$; $7.5 = 7.5$

When we compare case (i) with case (ii), the $x_{CM} = 2.5\,m$ from 3 kg mass could also be obtained by subtracting 4 m (the position of 3 kg mass) from 6.5 m, where the centre of mass was located in case (i)

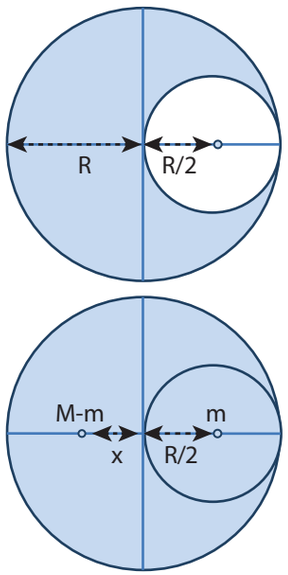
EXAMPLE 5.2

From a uniform disc of radius R, a small disc of radius $\frac{R}{2}$ is cut and removed as shown in the diagram. Find the centre of mass of the remaining portion of the disc.

Solution

Let us consider the mass of the uncut full disc be M. Its centre of mass would be at the geometric centre of the disc on which the origin coincides.

Let the mass of the small disc cut and removed be m and its centre of mass is at a position $\frac{R}{2}$ to the right of the origin as shown in the figure.



Hence, the remaining portion of the disc should have its centre of mass to the left of the origin; say, at a distance x. We can write from the principle of moments,

$$(M-m)x = (m)\frac{R}{2}$$

$$x = \left(\frac{m}{(M-m)}\right)\frac{R}{2}$$

If σ is the surface mass density (i.e. mass per unit surface area), $\sigma = \frac{M}{\pi R^2}$; then, the mass m of small disc is,

$$m = \text{surface mass density} \times \text{surface area}$$

$$m = \sigma \times \pi\left(\frac{R}{2}\right)^2$$

$$m = \left(\frac{M}{\pi R^2}\right)\pi\left(\frac{R}{2}\right)^2 = \frac{M}{\pi R^2}\pi\frac{R^2}{4} = \frac{M}{4}$$

substituting m in the expression for x

$$x = \frac{\frac{M}{4}}{\left(M-\frac{M}{4}\right)}\times\frac{R}{2} = \frac{\frac{M}{4}}{\left(\frac{3M}{4}\right)}\times\frac{R}{2}$$

$$x = \frac{R}{6}$$

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